Let’s look back at what you’ve accomplished so far in the design process and compare it with traditional counter design.

You designed a Counter smart contract that was similar in appearance to the code for a conventional counter. You may think it’s like a regular Java application that you run from the command line, but it is different.

When a smart contract code is deployed on a blockchain, it is accessible to anybody who has a defined identity on that blockchain. This identity could represent a participant (human or computer program) in Albany, New York, or in Bali, Indonesia; it is equally accessible as long as the participant has an account number and is connected to the same blockchain network. You may argue that this application is like any web application: a distributed system. But a distributed web application does not maintain a distributed immutable ledger. The blockchain does, and as a result, every participant records the same copy of the list of transactions that happened on the blockchain for this particular smart contract. This information is automatically logged with the agreement (consensus) of all the stakeholders on the network. Then the immutable ledger can be used to trace the provenance of the who, when, and what of the transactions. Thus, the distributed immutable ledger of the blockchain enables trust among unknown decentralized participant nodes, opening a whole world of opportunities for innovative Dapps, as you’ll see next.